

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Network and Cyber Security

Semester: Fall

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain the CIA Triad. How do confidentiality, integrity, and availability contribute to network security? 7
- b) What do you mean by computationally secure algorithm? Describe. 8
2. a) Describe the Data Encryption Standard (DES) algorithm. Explain its key generation process in detail. 7

OR

Explain the Advanced Encryption Standard (AES). Describe its encryption and decryption process in detail.

- b) Distinguish among DES, 2DES and 3DES in terms of their security parameters. 8
3. a) Why public key cryptography is needed? Using RSA, choose $p = 5$ and $q = 7$, encode the phrase "hello". 7
- b) Explain the Elgamal Cryptographic system with an example. 8
4. a) What is the Digital Signature Standard (DSS)? Explain its components and working principle. 7

OR

Explain key establishment techniques. Discuss the role of digital certificates in preventing Man-in-the-Middle attack.

- b) Explain authentication in symmetric key cryptography. Describe Message Authentication Code (MAC) and Message Digest (MD) with suitable examples. 8
5. a) What do you mean by SSL? Explain SSL handshake protocol. 7
- b) Explain Pretty Good Privacy (PGP). Describe its working principle and security services provided. 8

6. a) What is Server-Side Request Forgery (SSRF)? Explain its causes, impacts, and mitigation techniques. 7
- b) Explain the OWASP Top 10 vulnerabilities. Discuss their significance in securing modern web applications. 8
7. Write short notes on: (Any two) 2×5
- a) Elliptic curve cryptography
- b) IPsec Protocol
- c) Cryptographic Failures